

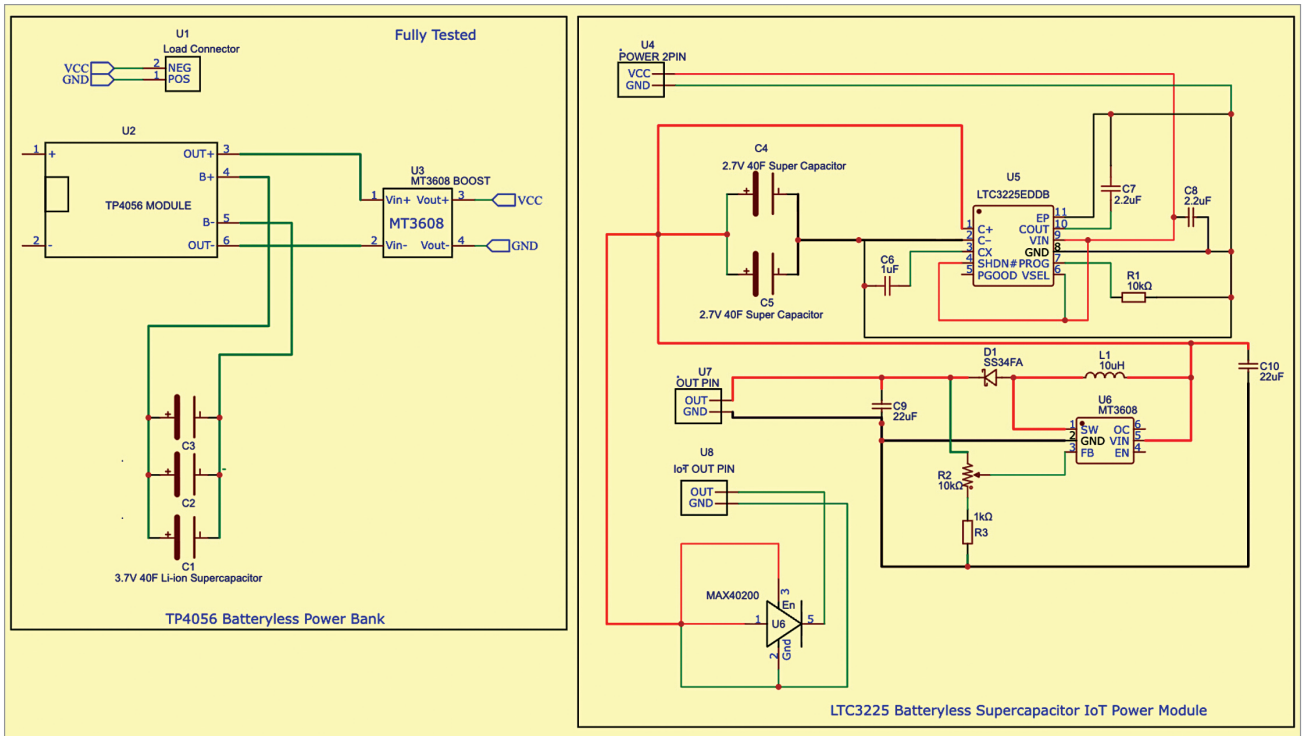


Fig. 2: Batteryless power bank circuit

similarly to a lithium battery electrode, creating a hybrid structure. This enables LICs to combine high power delivery with improved energy density, making them

well-suited for short-term backup applications in IoT systems. Fig. 1 presents the assembled prototype of the supercapacitor-based backup power module. The top-left image shows the prototype powering an IoT device without a conventional battery. The internal view highlights a stacked perfboard structure with two large supercapacitors mounted above the main PCB and a blue LED indicating power status. The top-right image shows a simple power bank design using a hybrid lithium-ion supercapacitor, where a compact micro-USB charging module charges the supercapacitors and manages the backup power system.

The system utilises lithium-ion supercapacitors (hybrid supercapacitors), which can store charge and exhibit very low self-discharge compared to conventional capacitors.

The components required to build this system are listed in the Bill of Materials table.

Circuit design

The system includes two circuit implementations of a supercapacitor-based power bank for short-duration backup power. These circuits enable electronic devices to complete critical operations such as saving data, transmitting final information, or executing safe shutdown procedures.

First design - TP4056-based batteryless power backup. In the first design, the TP4056 is used to charge the supercapacitor, while the MT3608 boost DC-DC converter generates the required output voltage.

The MT3608 provides an adjustable output in the range of approximately 5V to 25V DC, with higher voltages achievable under light-load conditions.

However, this approach is not recommended because the TP4056 is designed specifically for lithium-ion batteries. Using it to charge supercapacitors may result in reduced efficiency and suboptimal charging behaviour. Despite this, testing confirms that the circuit can

BILL OF MATERIALS

Name	Designator	Quantity	Manufacturer
3.7V, 40F Li-ion supercapacitor	C1, C2, C3	3	
2.7V, 40F Li-ion supercapacitor	C4, C5	2	
1μF	C6	1	Samsung
2.2μF	C7, C8	2	Samsung
22μF	C9, C10	2	Samsung
SS34FA	D1	1	Onsemi
10μH	L1	1	Sxn
10kΩ	R1	1	Yageo
10kΩ (potmeter)	R2	1	Amphenol
1kΩ	R3	1	Yageo
Load connector	U1	1	
TP4056 module	U2	1	ADI
MT3608	U3, U6	2	
Power 2-pin	U4	1	
LTC3225EDDB	U5	1	
Out pin	U7	1	
IoT out pin	U8	1	
MAX40200	U9	1	